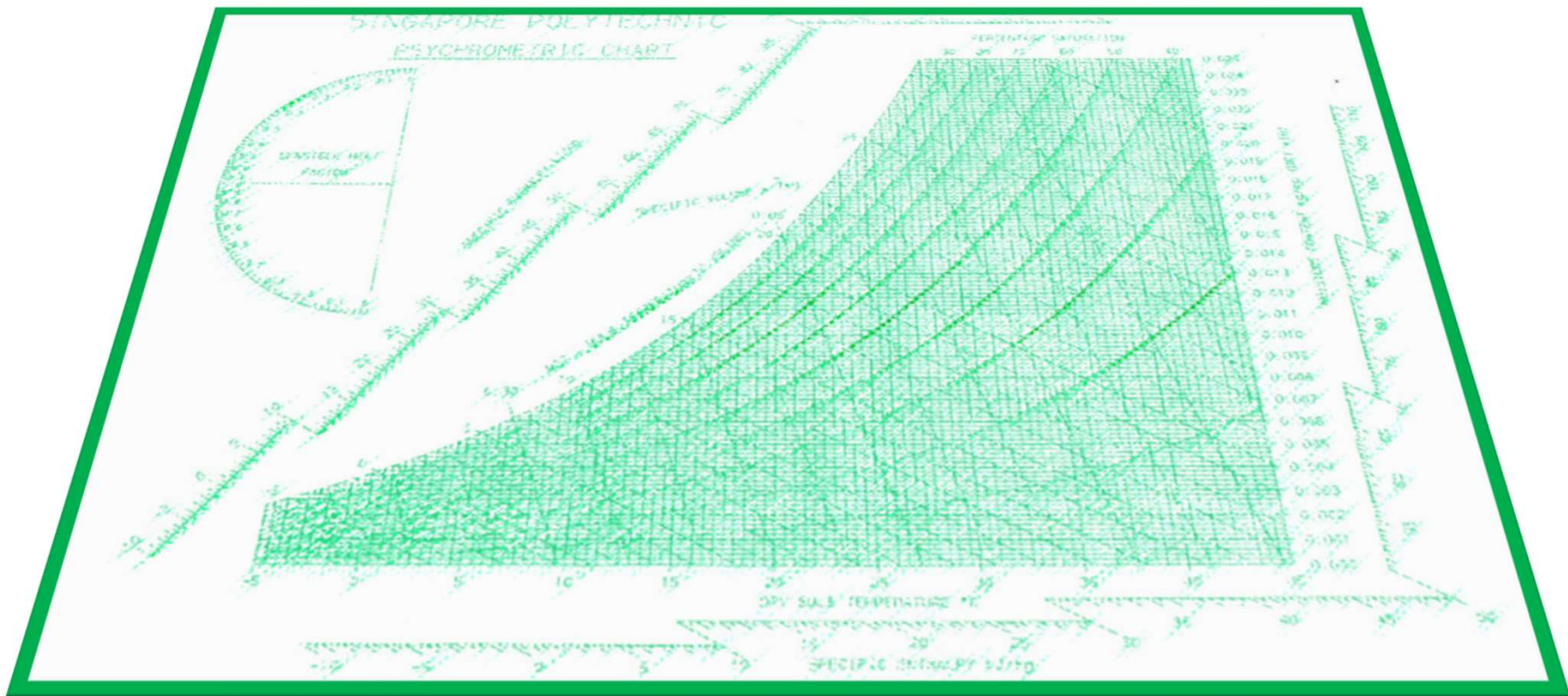
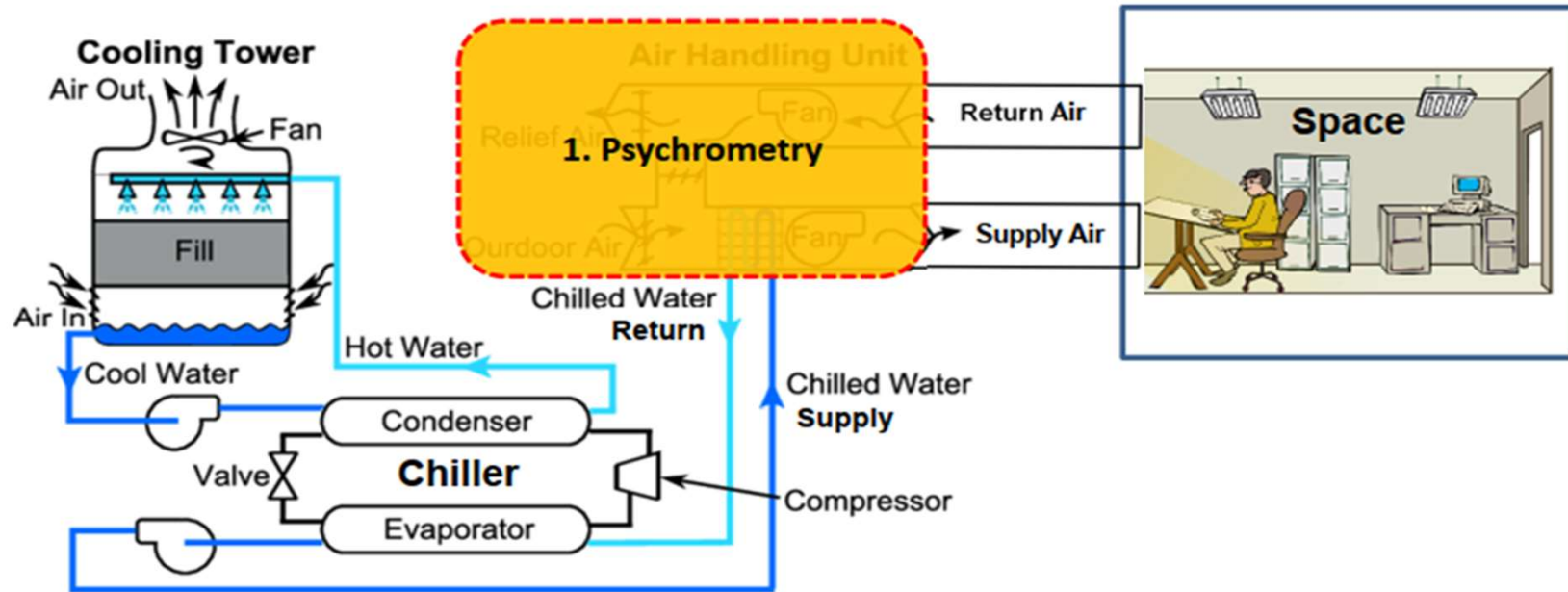


Psychrometry



Presented By Sandar Moe

Psychrometry

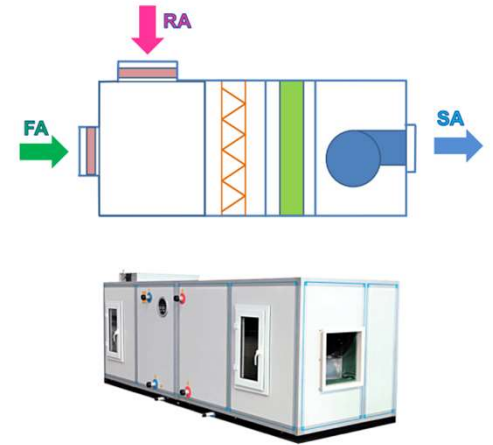


AIR SYSTEM :

THREE Different Way Distribute to the Room

–Constant Air Volume (CAV) System , Variable Air Volume (VAV) System , Fan Coil System

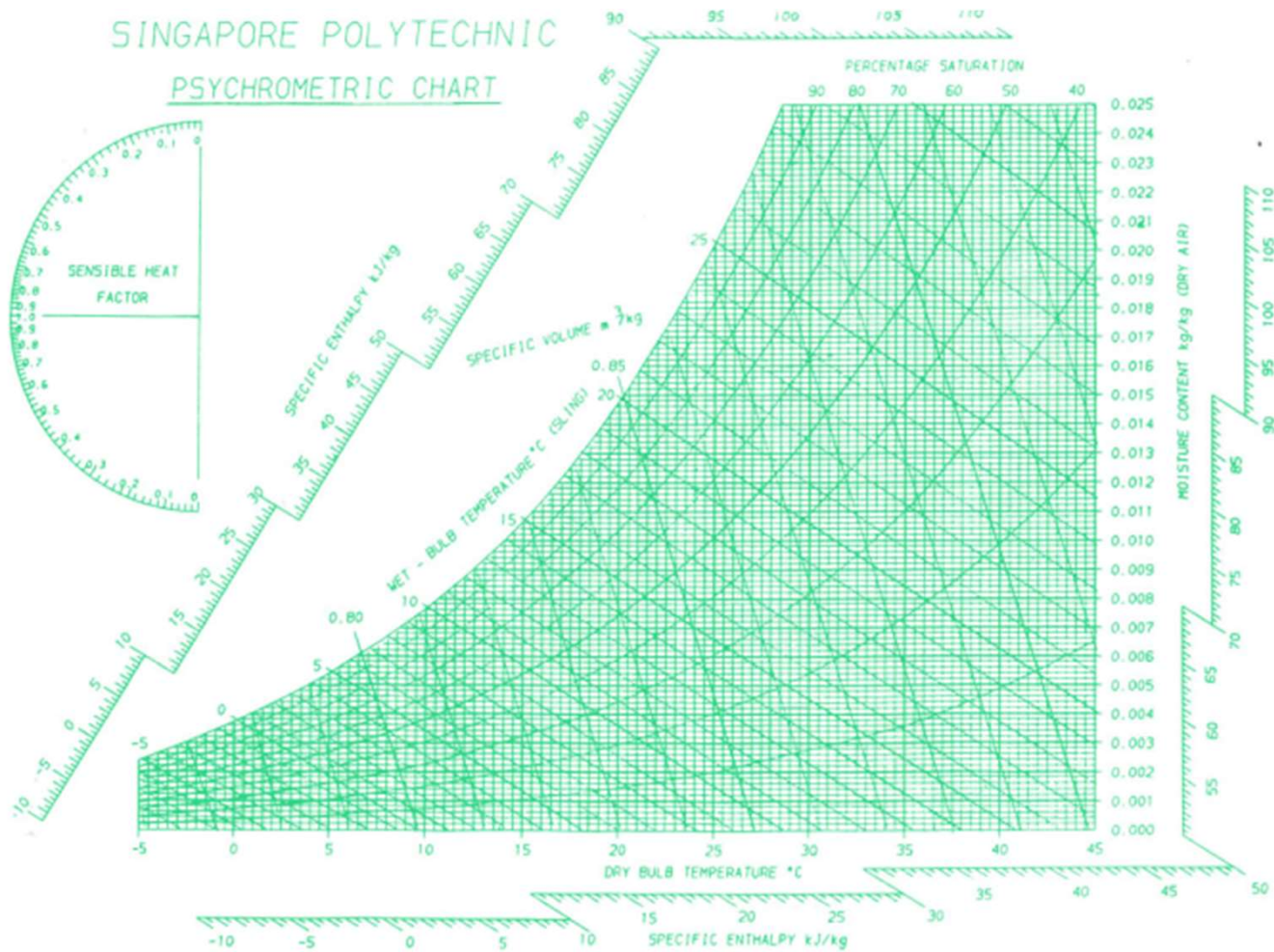
Psychrometry



Understanding about Psychrometry Process

- Psychrometry is the relationship that exist within a mixture of air and water vapour .
- When designing an air conditioning system , the temperature and moisture content of the air to be conditioned & the air conditioning task can be determined .
- The generally Psychrometric chart displays several physical properties of air over a broad range of conditions .

Psychrometry



Psychrometry

Basic Psychrometric Terms

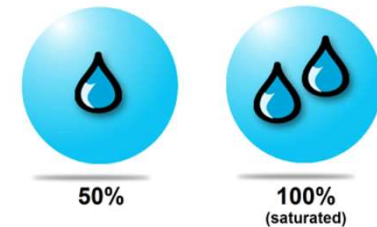
- Dry Bulb Temperature : The temperature of the air measured by an Ordinary thermometer.
- Wet Bulb Temperature : This is the temperature of the air as indicated by a thermometer which has a wet bulb (Bulb covered with Wet Cloth or Gauze) over which air flows. If the air is not saturated , the water will evaporate round the wet bulb , which will cause a fall in temperature of the bulb.
- Dew Point Temperature : This is the temperature at which water Vapour would start to condense out of the air if the air were to be cooled at constant pressure . When the air is at the dew point temperature , It is holding all the moisture it can hold at that temperature . Condensation occurs at Dew Point Temperature .
- Specific Humidity or Moisture Content (w) : This is the mass of water vapour in moist air per Kg of its dry air content (Kg/Kg Dry Air) . The actual weight of water in an air –water vapour mixture . In other words , if one Kg of Air were wrung completely dry , comparing the weight of the water to the weight of the dry air would yield its specific humidity .

Psychrometry

- Relative Humidity (%RH) : The **RATIO** of the mass of water vapour .

For example, if the relative humidity of the air is 50%, it contains one-half the amount of moisture possible at the existing dry-bulb temperature.

$$\text{Relative Humidity} = \frac{\text{Amount of moisture that a given amount of air is holding}}{\text{Amount of moisture that a given amount of air can hold}}$$



Psychrometry

Specific Volume (v)

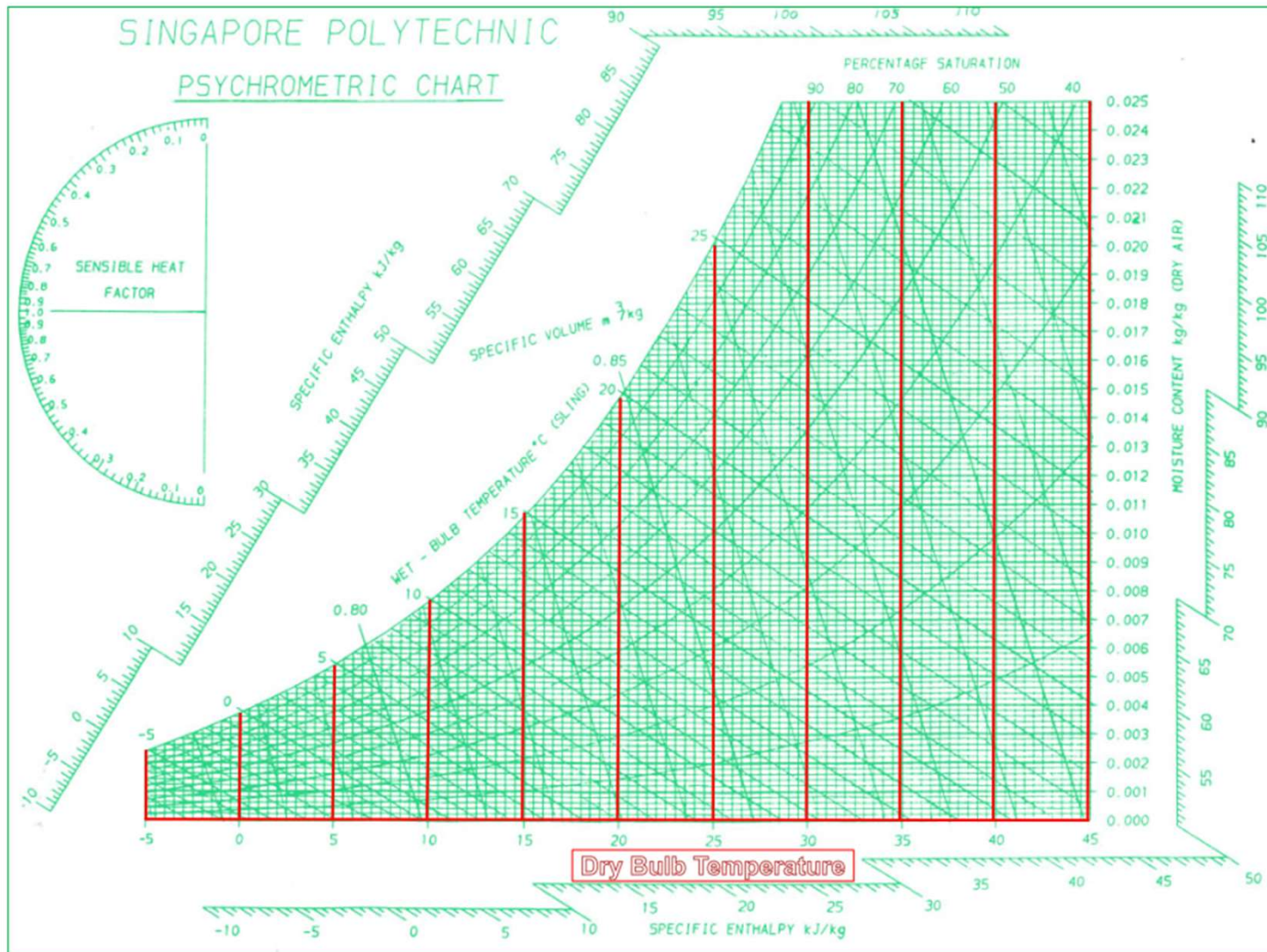
This is the volume of moist air per kg of dry air content.

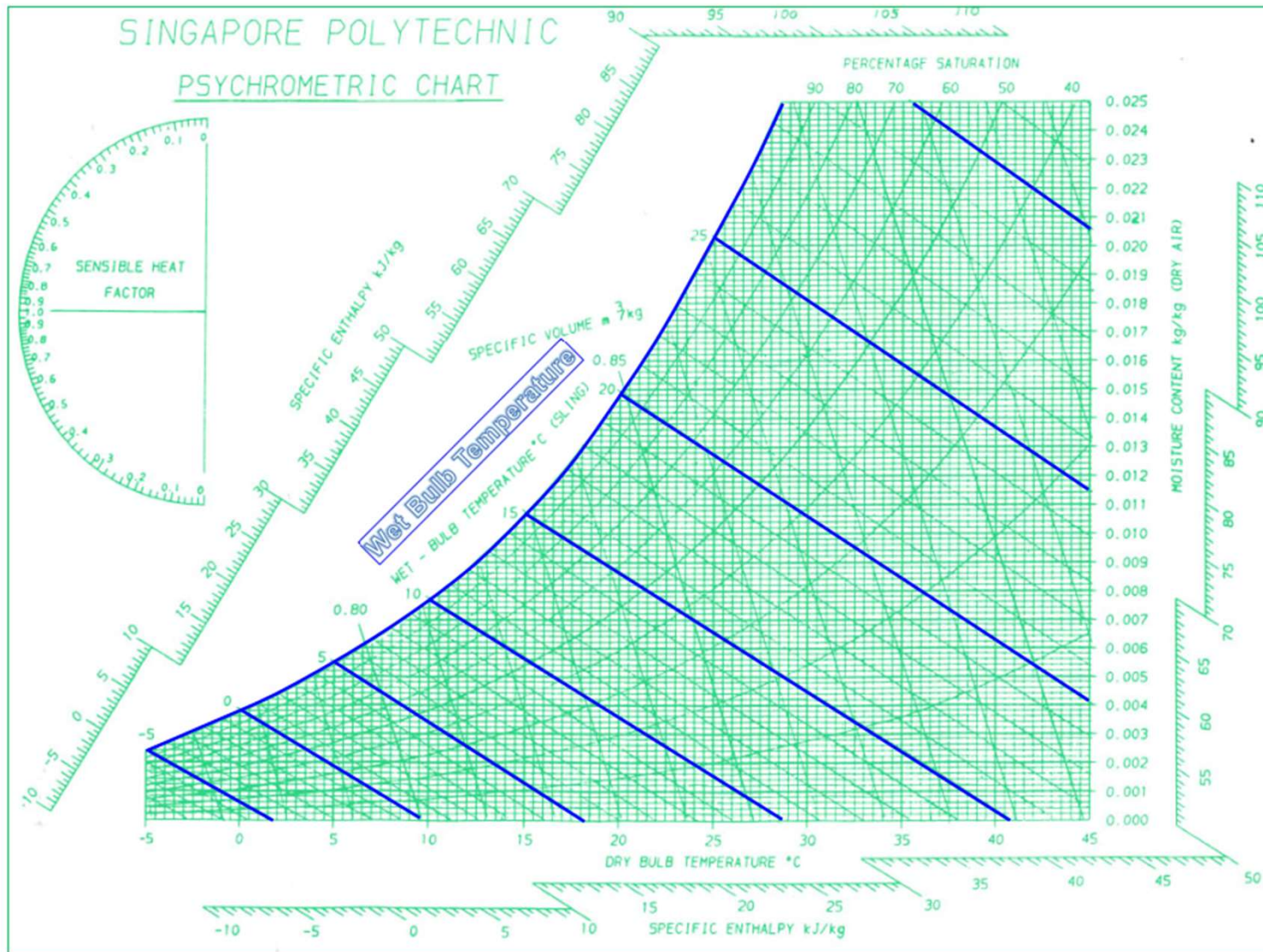
$$v = \frac{\dot{V}}{\dot{m}} \quad \left(\frac{m^3}{kg} \right)$$

Specific Enthalpy (h)

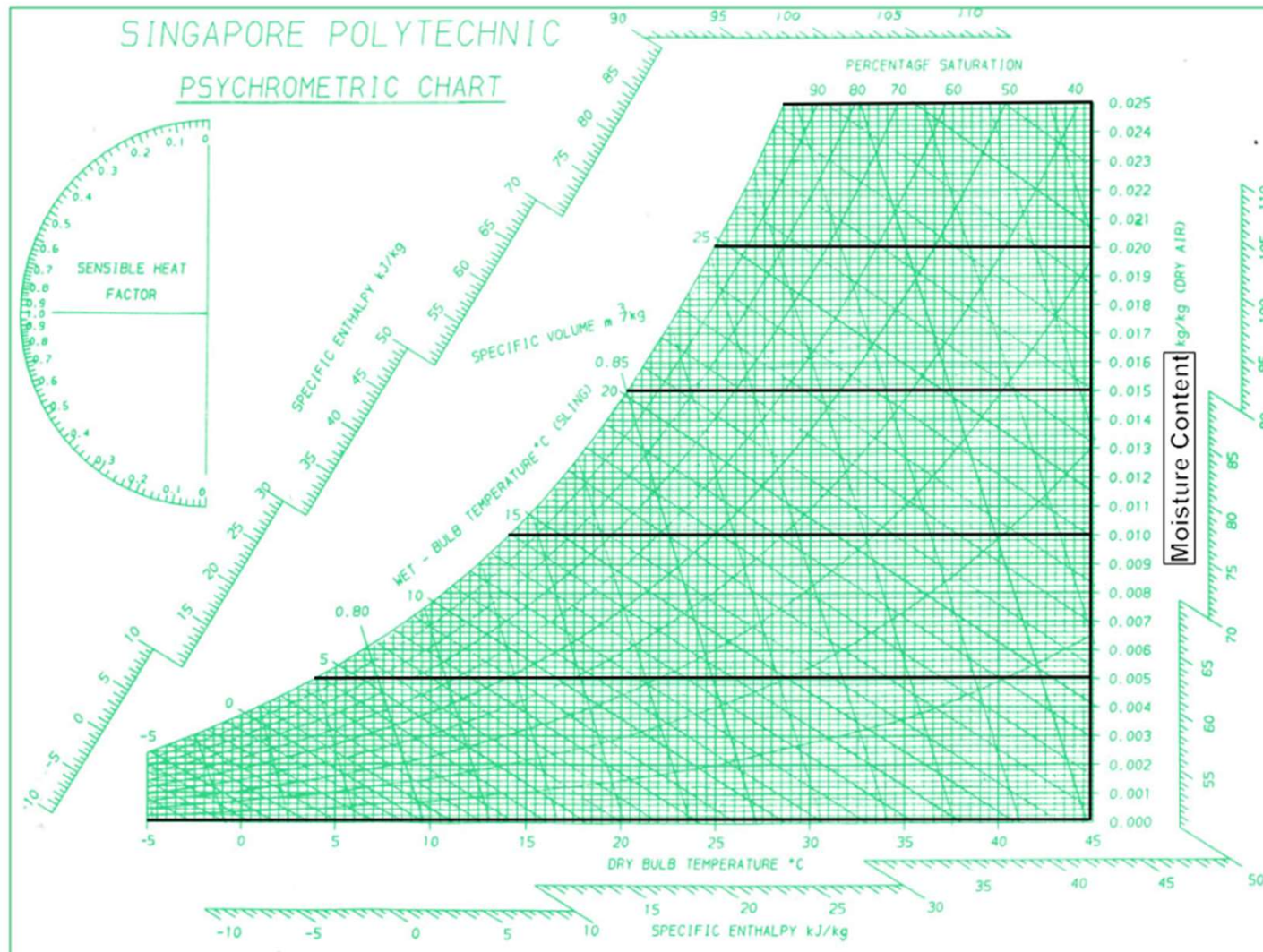
This is the heat content in the dry air and the moisture content in the air per kg of air content. The specific enthalpy is arbitrarily chosen as zero at 0°Cdb temperature (Check psychrometric chart).

Psychrometry

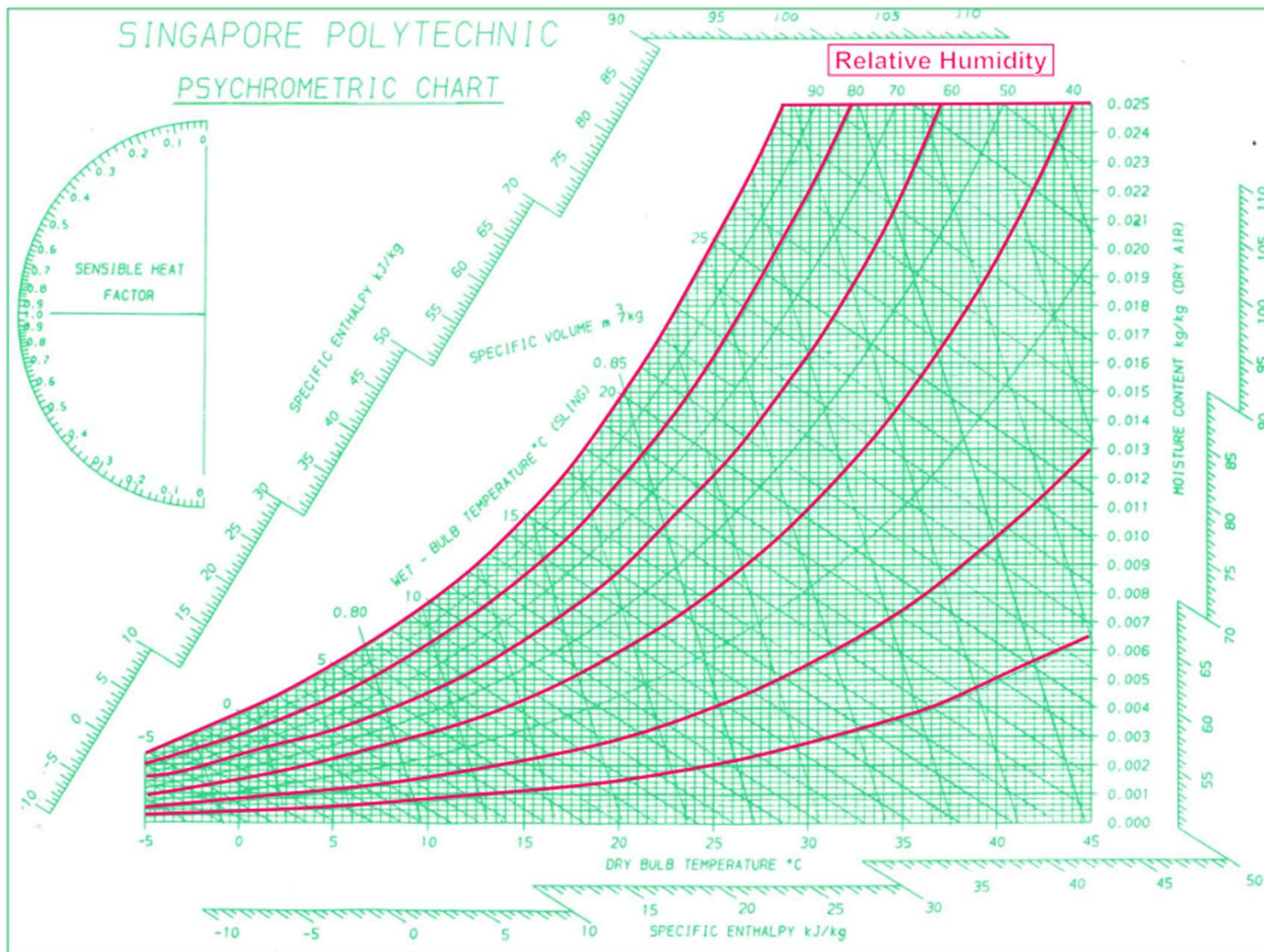




Psychrometry

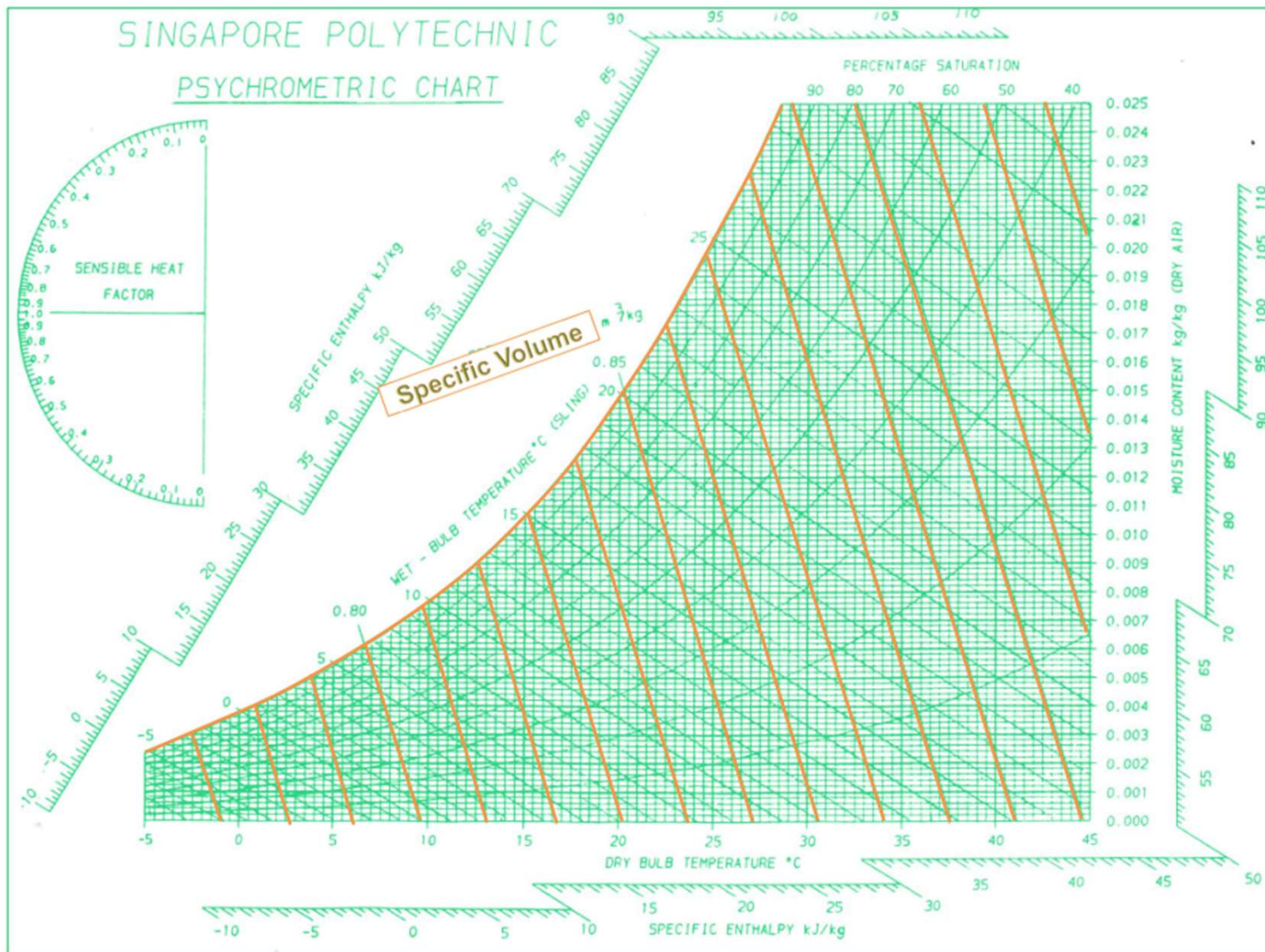


Psychrometry



Psychrometry

SINGAPORE POLYTECHNIC
PSYCHROMETRIC CHART

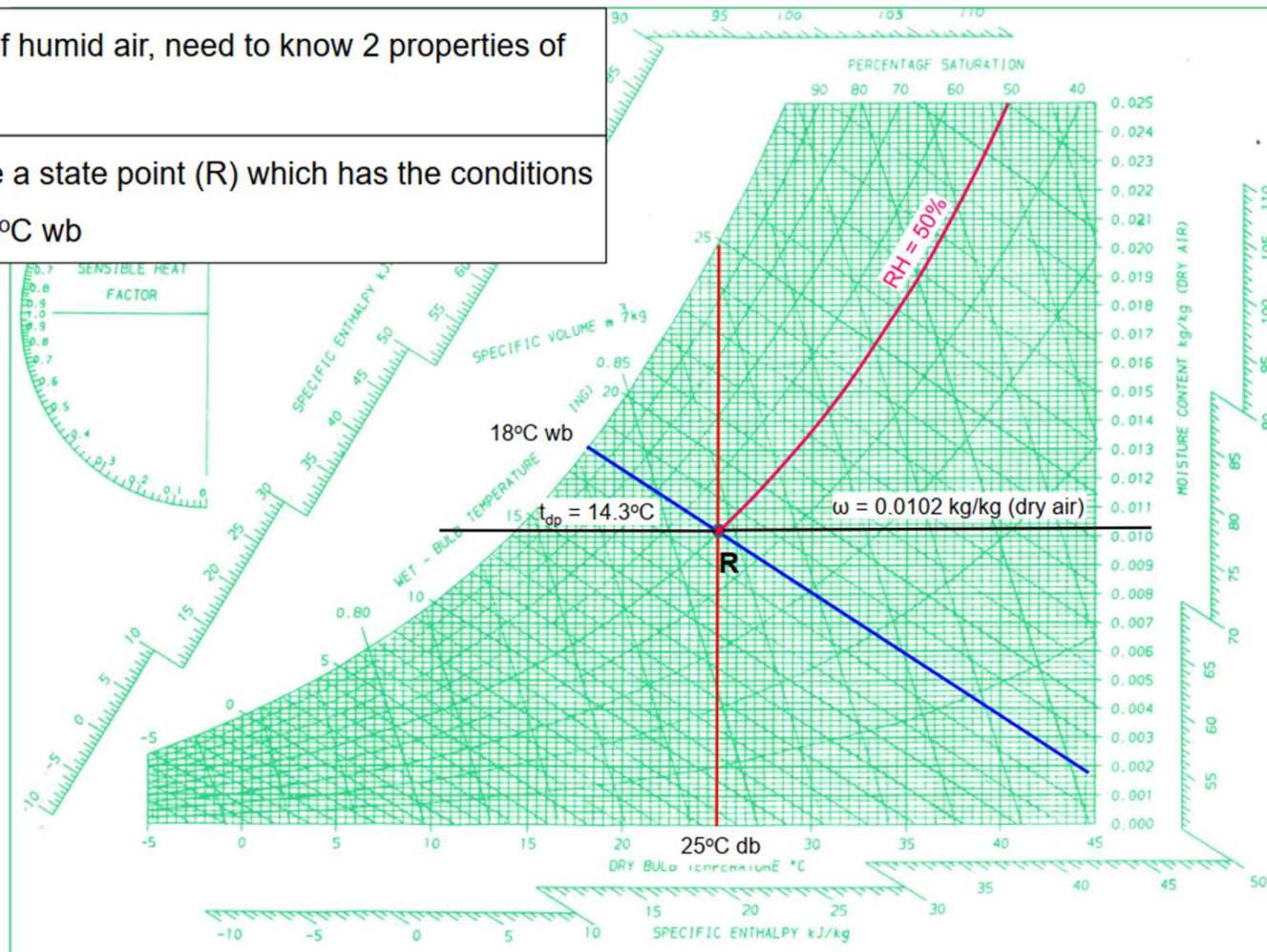


Psychrometry

Psychrometry

To locate a point of humid air, need to know 2 properties of that humid air.

Example: to locate a state point (R) which has the conditions of 25°C db and 18°C wb



In the Psychrometric Processes , The AIR is either HEATED or COOLED and MOISTURE is either added or removed . SENSIBLE HEAT and LATENT HEAT

SENSIBLE HEAT (Q_{sh}) : Heat that cause a change in temperature but not in Moisture Content. The Heat can be detected by human senses

$$Q_{SH} = \dot{m} \cdot C_p \cdot (\Delta t)$$

Where Q_{SH} = sensible heat in kW
 $\dot{m}$ = mass flow rate of air in kg/s
 C_p = specific heat capacity of air, 1.01 kJ/kgK
 Δt = change in temperature of the air in K

LATENT HEAT (Q_{LH}) : Heat that cause a change in physical state (Phase) but not in Temperature.

Eg. Boiling of water which change the liquid to vapour . The Moisture content of the air changes but not its temperature .

$$Q_{LH} = \dot{m} \cdot H_L \cdot (\Delta\omega)$$

Where Q_{LH} = latent heat in kW

$\dot{m}$ = mass flow rate of air in kg/s

H_L = latent heat of vaporisation of water, 2450 kJ/kg

$\Delta\omega$ = change in moisture content of the air in kg/kg

Total Heat (Q_{TH})

This is the total heat due to changes in both sensible heat and latent heat.

$$Q_{TH} = Q_{SH} + Q_{LH}$$

$$Q_{TH} = \dot{m} \cdot C_p \cdot (\Delta t) + \dot{m} \cdot H_L \cdot (\Delta \omega)$$

$$Q_{TH} = \dot{m} \cdot (\Delta h)$$

Where

Q_{TH} = total heat in kW

Q_{SH} = sensible heat in kW

Q_{LH} = latent heat in kW

$\dot{m}$ = mass flow rate of air in kg/s

C_p = specific heat capacity of air, 1.01 kJ/kgK

Δt = change in temperature of the air in K

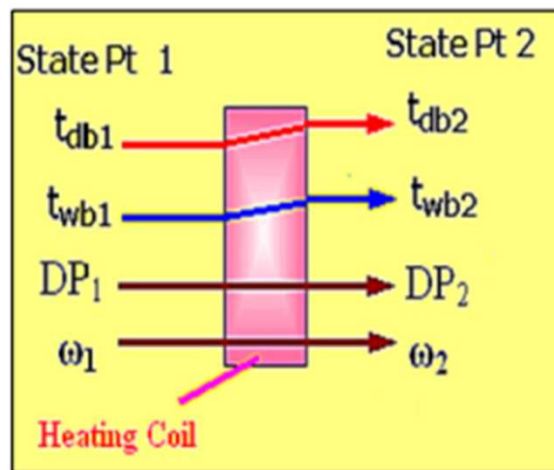
H_L = latent heat of vaporisation of water, 2450 kJ/kg

$\Delta \omega$ = change in moisture content of the air in kg/kg

Δh = change in specific enthalpy of the air in kJ/kg

Sensible Heating

When air flows through a steam or hot water coil or electric heater, this air is heated and its temperature and heat content increases while the moisture content (ω) remains unchanged. Sensible heating is represented by a horizontal line on the psychrometric chart.



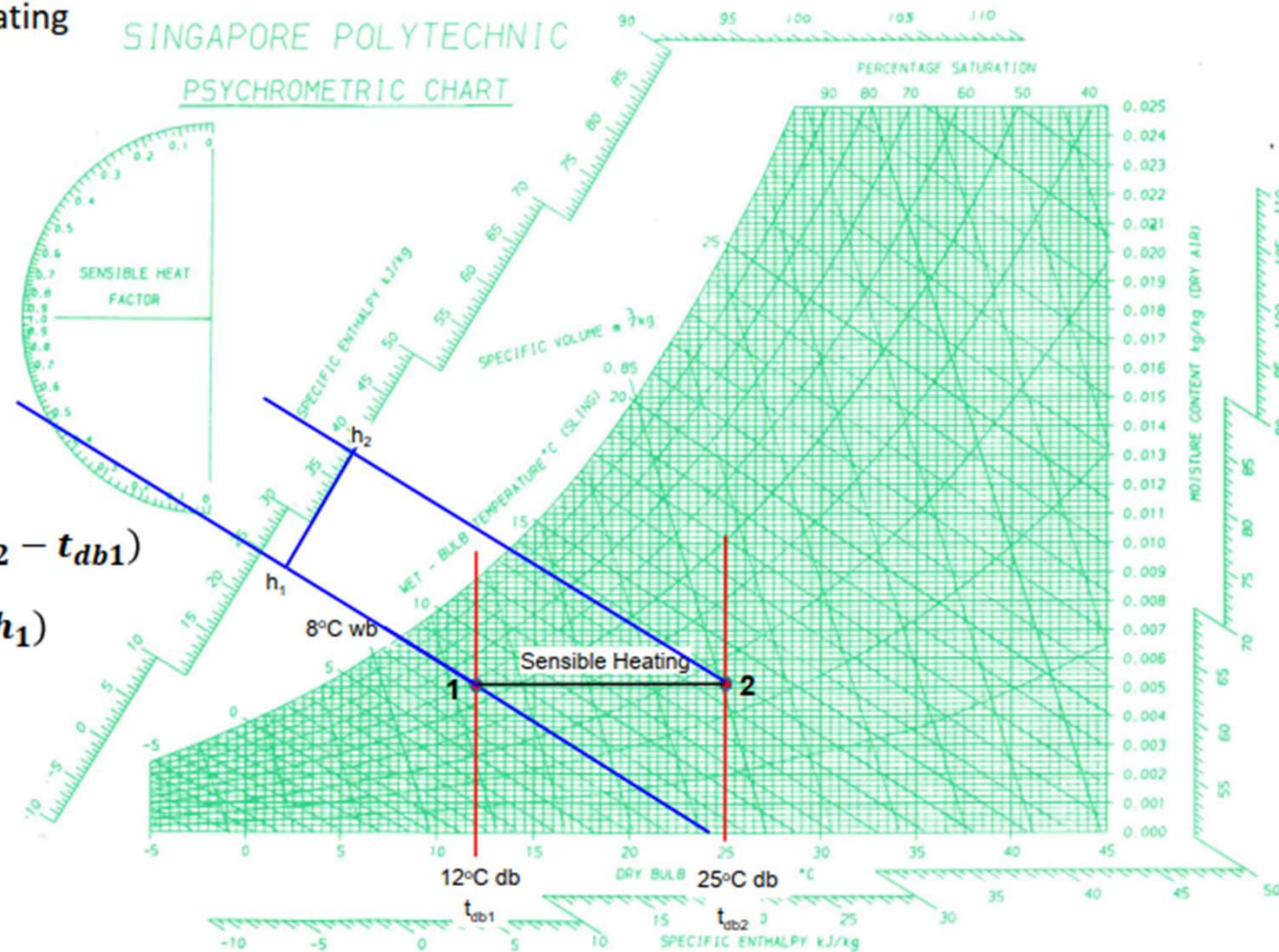
Sensible Heating

Pt 1: 12°C db /
8°C wb

Pt 2: 25°C db

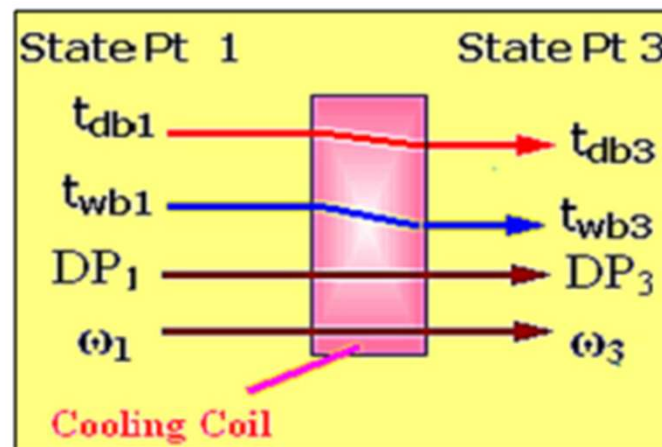
$$Q_{12} = \dot{m} C_p (t_{db2} - t_{db1})$$

$$Q_{12} = \dot{m} (h_2 - h_1)$$



Sensible Cooling

The air flowing through a cooling coil is cooled to a temperature above its dew point so that no moisture is removed. A horizontal on a psychrometric chart represents this process.



Psychrometry

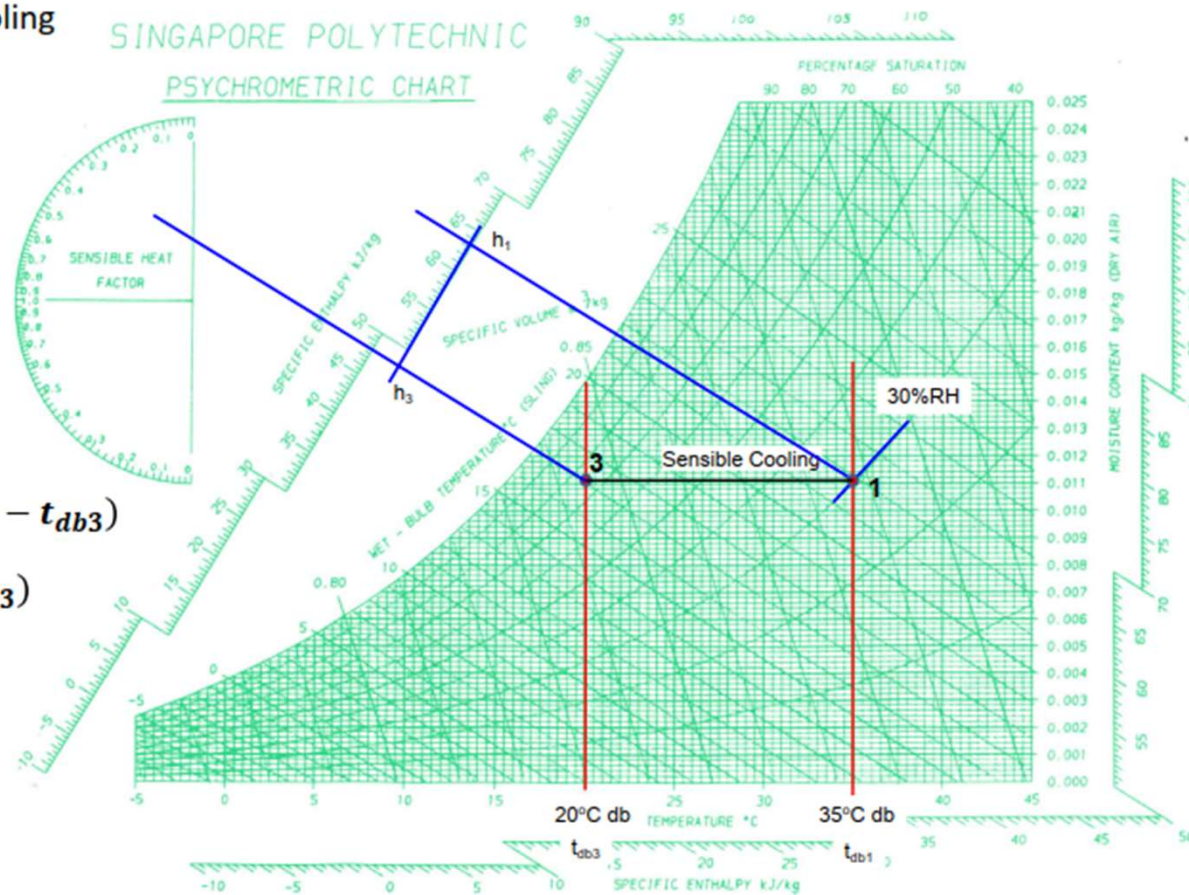
Sensible Cooling

Pt 1: 35°C db /
30%RH

Pt 3: 20°C db

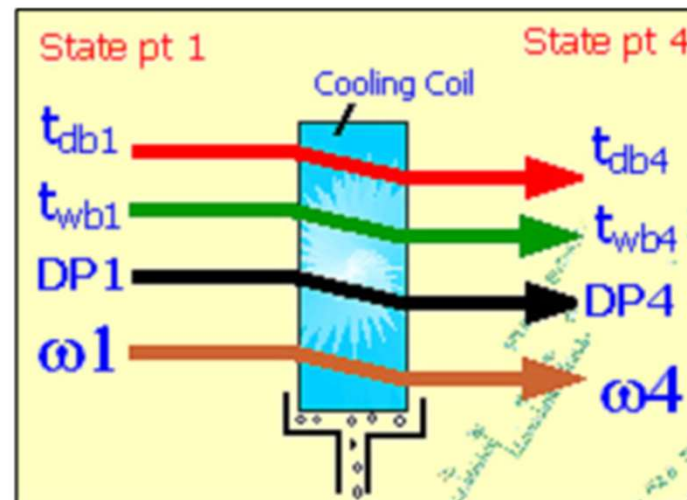
$$Q_{13} = \dot{m}C_p(t_{db1} - t_{db3})$$

$$Q_{13} = \dot{m}(h_1 - h_3)$$



Cooling & Dehumidification

The air flowing through a cooling coil is cooled to a temperature below its dew point. **Both** sensible and latent heats are removed from the air.



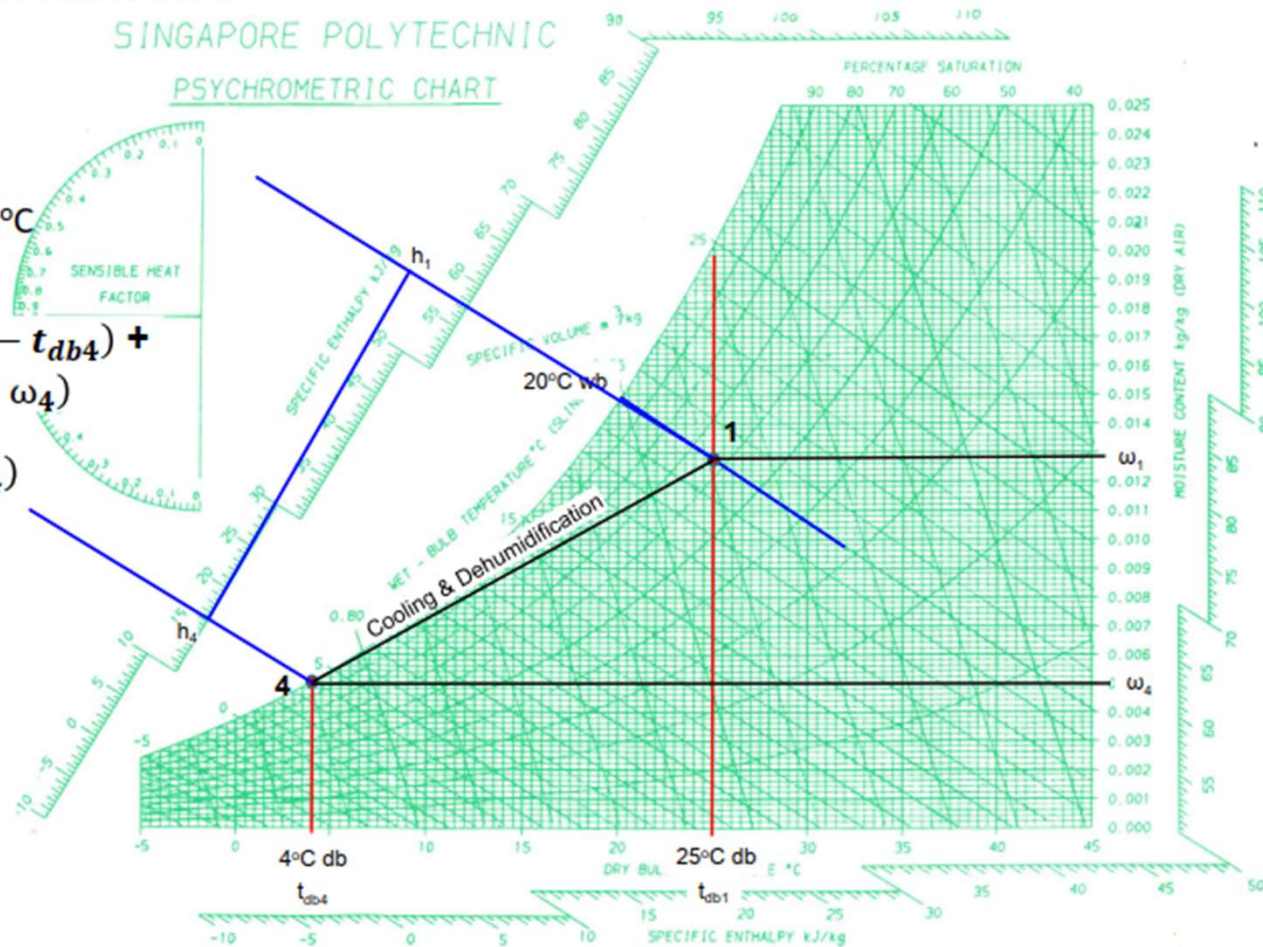
Cooling & Dehumidification

Pt 1: 25°C db /
20°C wb

Pt 4: Saturated 4°C
db (or) 4°C wb

$$Q_{14} = \dot{m}C_p(t_{db1} - t_{db4}) + \dot{m}H_L(\omega_1 - \omega_4)$$

$$Q_{14} = \dot{m}(h_1 - h_4)$$



Sensible Heat Factor (SHF) / Room Sensible Heat Factor (RSHF)

It is defined as the ratio of the sensible heat to the total heat of the air during a process. The range of SHF values is between 0 and 1. If a cooling process involves the removal of 80% of sensible heat and 20% of latent heat, then the SHF is 0.8. Hence, by definition

$$\text{SHF} = \frac{SH}{SH + LH}$$

$$SH = \dot{m}C_p(t_2 - t_1)$$

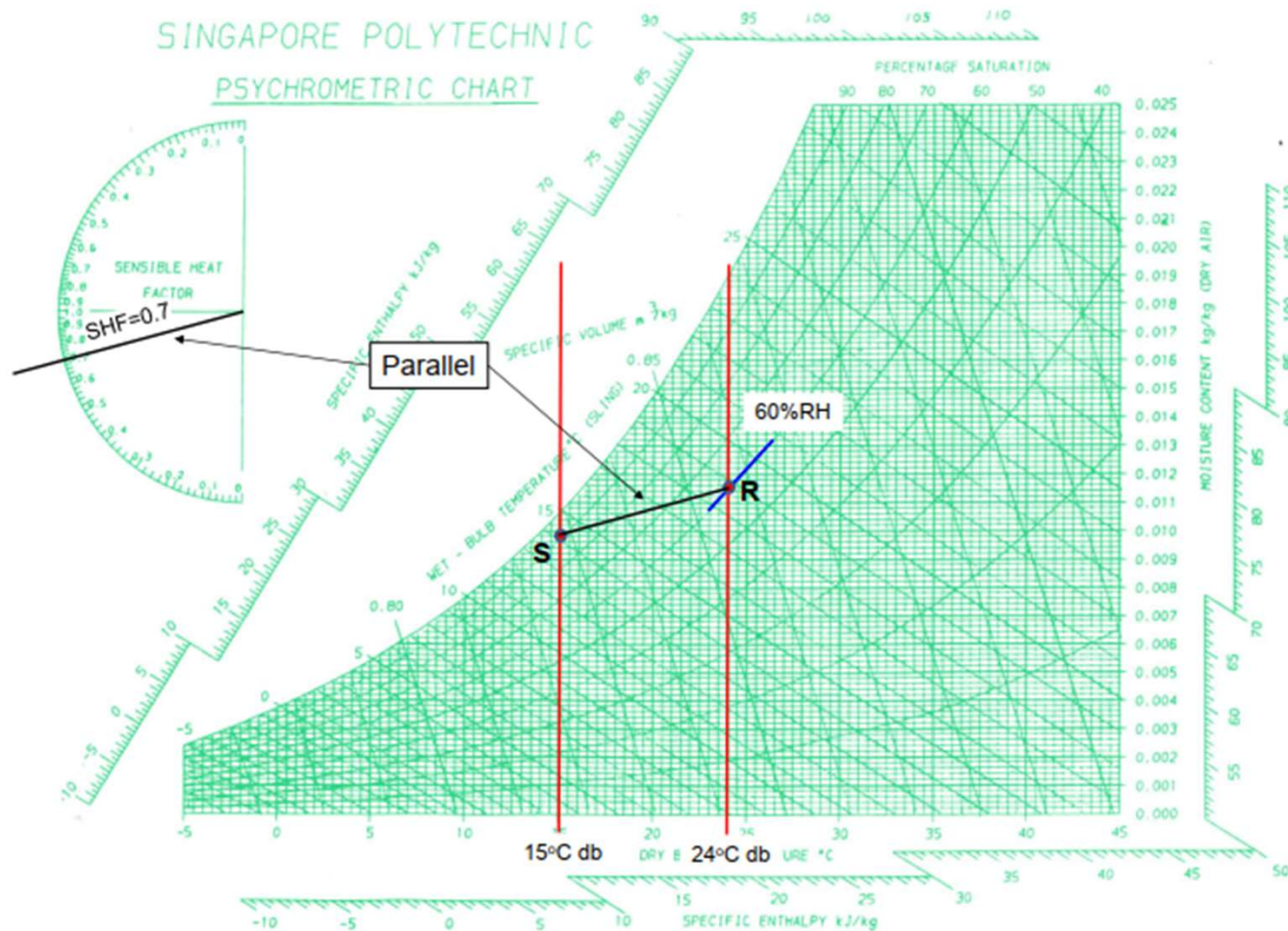
$$LH = \dot{m}H_L(\omega_2 - \omega_1)$$

Sensible Heat Factor (SHF) / Room Sensible Heat Factor (RSHF)

SHF: 0.7

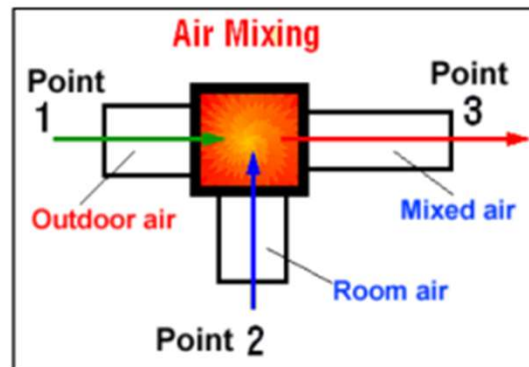
**Room: 24°C db /
60% RH**

Supply: 15°C db



Air Mixing Process

In an air mixing process as shown below, air stream 1 mixes with air stream 2 to give air stream 3. An example of air mixing in air-conditioning is the mixing of the return air with the outdoor air. The process can be plotted on a psychrometric chart and the air mixture condition can be determined. In the air mixing process, the total heat content, moisture content and mass of the air before and after the process is the same,



$$\dot{m}_3 = \dot{m}_1 + \dot{m}_2$$

$$\frac{\dot{m}_1}{\dot{m}_3} = \frac{h_2 - h_3}{h_2 - h_1}$$

$$\frac{\dot{m}_1}{\dot{m}_3} = \frac{t_2 - t_3}{t_2 - t_1}$$

$$\frac{\dot{m}_1}{\dot{m}_3} = \frac{\omega_2 - \omega_3}{\omega_2 - \omega_1}$$

Air Mixing Process

m_1 to $m_2 \rightarrow 1:2$

Pt1: 33°C db / 28°C wb
wb

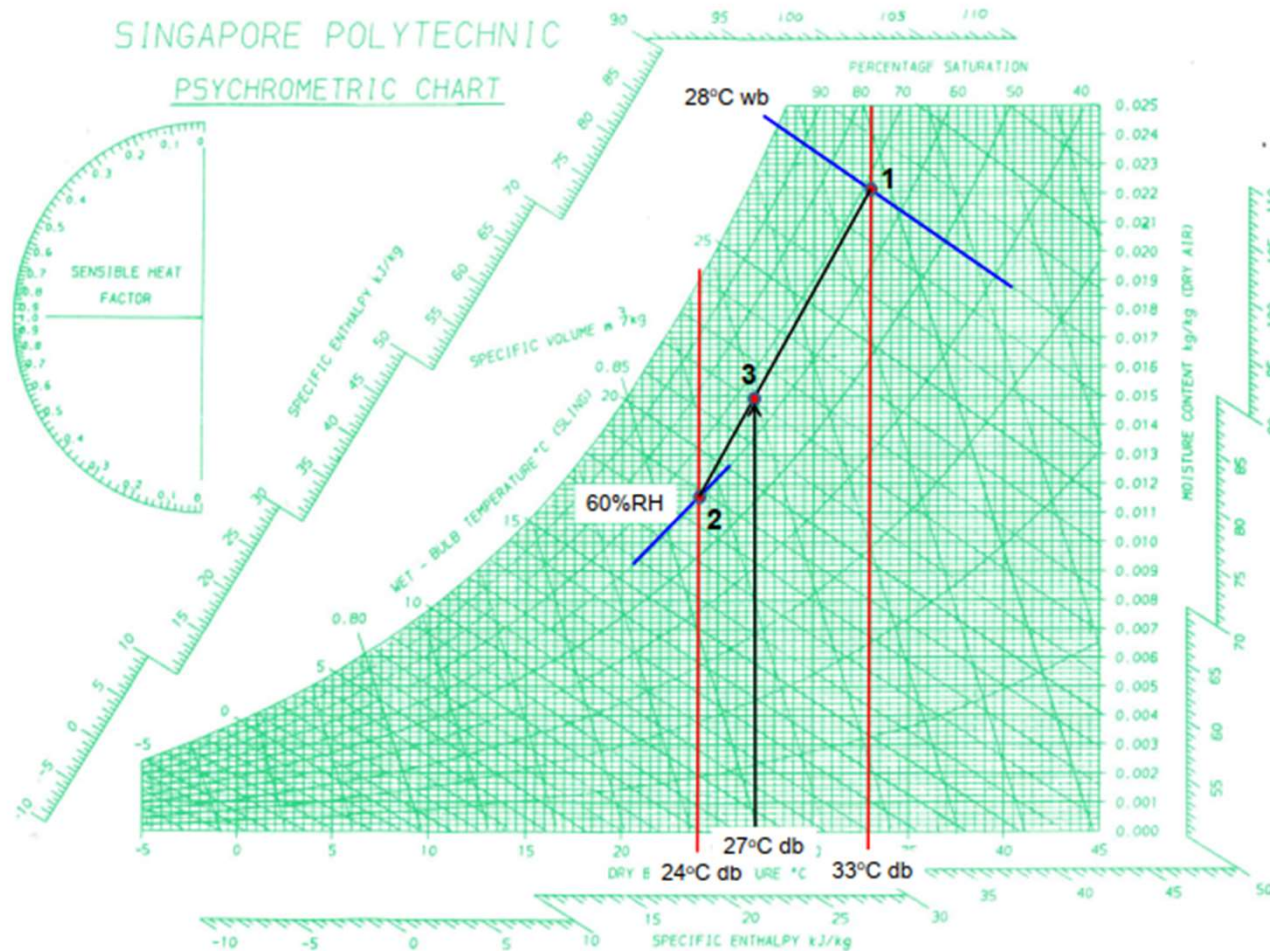
Pt2: 24°C db / 60% RH
RH

$m_1/m_3 \rightarrow 1/3$

$$\frac{\dot{m}_1}{\dot{m}_3} = \frac{t_2 - t_3}{t_2 - t_1}$$

$$\frac{1}{3} = \frac{24 - t_3}{24 - 33}$$

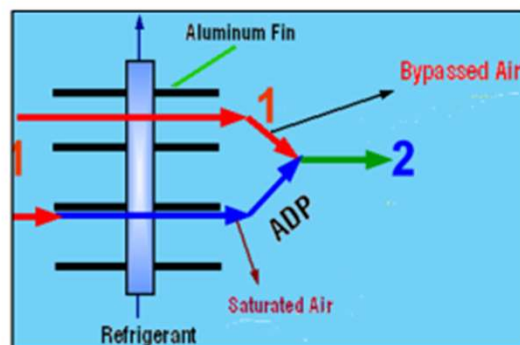
$$t_3 = 27$$



Apparatus Dew Point (ADP) and Bypass Factor (BF)

When air flows pass the cooling coil, it will come in contact with the surface of the coil. ADP is the temperature to which the air that contacts the surface of the cooling coil is cooled. If all the air passing over the coil contacts the coil surface, the air will leave at a temperature equal to ADP. Since ADP is on the saturated line, this air would be saturated.

In practice, when air at condition "1" enters a cooling coil as shown in the figure, not all air comes into contact with the coil surface. Only part of it will be in contact with the coil surface and leave with condition "ADP". The remaining part of the air will go straight through and is not affected by the coil, thus leaving the coil at the entering condition "1". The two parts then combine again to form condition "2" on leaving the coil.



Apparatus Dew Point (ADP) and Bypass Factor (BF)

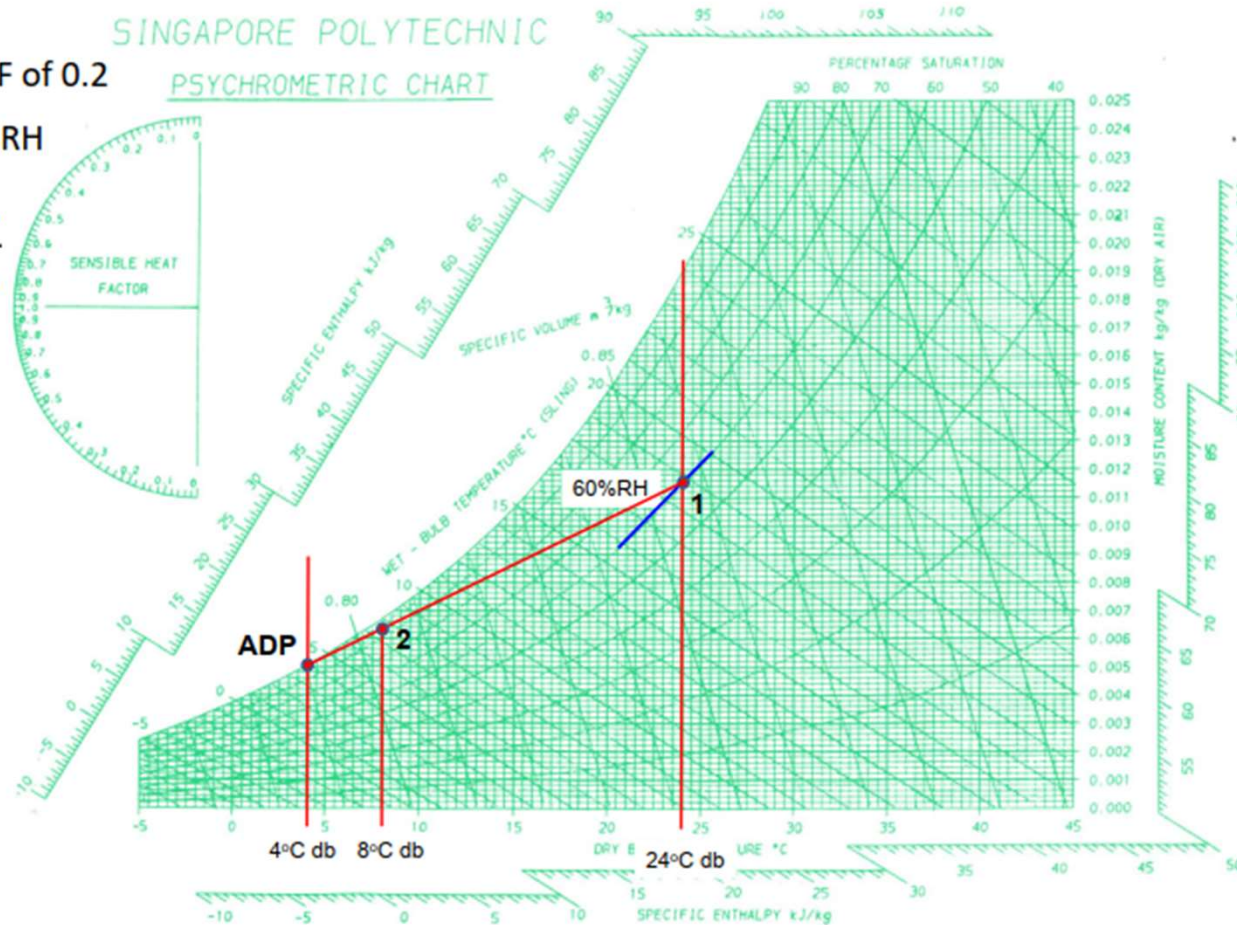
ADP: 4°C db with BF of 0.2

Pt1: 24°C db / 60% RH

$$BF = \frac{t_2 - t_{ADP}}{t_1 - t_{ADP}}$$

$$0.2 = \frac{t_2 - 4}{24 - 4}$$

$$t_2 = 8$$



THANK YOU

Q & A